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Introduction to PID Controller and Model Predictive Control in Engineering Systems

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Abstract— This paper presents an exploration of Proportional-Integral-Derivative (PID) controllers and Model Predictive Control (MPC) within the realm of autonomous systems, emphasizing their pivotal roles in enhancing the operational efficacy and intelligence of robotic systems. The initial segment delves into the foundational aspects of PID control, illustrating its ubiquitous presence in regulating diverse mechanisms from household appliances to advanced automotive systems. Through historical context, the evolution of control systems is traced back to seminal inventions like the flyball governor, demonstrating the progression towards automated control in managing steam engine power outputs. Subsequently, the discussion transitions to Model Predictive Control (MPC), a sophisticated control strategy that transcends traditional PID methodologies by integrating predictive models of the system's future behavior. MPC's efficacy is highlighted in its application to autonomous vehicles, showcasing its capability to execute complex maneuvers such as highway speed regulation, precise parking, and obstacle avoidance through predictive and adaptive control mechanisms. This is achieved by formulating a control problem as an optimization problem over a future time horizon, thereby allowing for the anticipation of future events and the adjustment of control actions accordingly. The narrative underscores the critical distinction between sensing, decision-making, and actuation layers within robotic systems, situating MPC within the actuation layer, albeit with overlaps into decision-making due to its predictive nature. Through a series of structured experiments and assignments, the paper elucidates how MPC can address challenges inherent in dynamic environments, such as variable loads in elevators and navigation in autonomous driving, illustrating the versatility and robustness of MPC in contrast to conventional control strategies. In essence, this study aims to articulate the transformative potential of PID controllers and MPC in advancing autonomous systems, underlining the importance of predictive control in achieving precise, reliable, and adaptable automation. By examining the operational principles, applications, and implications of these control strategies, the paper contributes to the broader understanding and appreciation of their significance in the evolving landscape of robotics and automated systems.

I. INTRODUCTION

In the introductory section of this research, we explore the intricate world of control systems, with a special emphasis on Proportional, Integral, and Derivative (PID) controllers. Our aim is to demystify the principles and mechanisms that govern these systems, paving the way for a deeper comprehension of their design and practical applications. At the heart of any effective control system lies the controller, an

essential component tasked with regulating a system's behavior to achieve a specified output. This fundamental aspect of control systems theory is crucial for understanding how various inputs and outputs interact within a system to produce desired outcomes.

Through a broad examination, we establish the relevance of control systems in numerous domains, highlighting their capability to manage and optimize system dynamics in diverse settings. The core principle of control systems involves a dynamic response to discrepancies between actual states and desired states, often referred to as the error. This response is facilitated by the implementation of feedback mechanisms, enabling real-time adjustments to system inputs based on the observed error. Such adjustments ensure the alignment of the system's actual output with its intended output, thereby achieving the desired control objective.

The discourse extends to the inputs and outputs characteristic of control systems, where outputs are defined as observable and measurable parameters that are crucial for the system's intended function. Inputs, conversely, are actions or external factors that influence the system's output. The interplay between input and output is foundational to control theory, illustrating the system's capacity to modify its behavior in pursuit of a specified goal.

This exploration sets a solid groundwork for discussing the calculus-based design and implementation of PID controllers. By dissecting the theoretical underpinnings and practical implications of these controllers, the paper aims to furnish readers with a comprehensive understanding of their operational efficacy and versatility. Furthermore, the introduction serves as a precursor to a discussion on Model Predictive Control (MPC), a sophisticated control strategy that builds upon the principles of PID control by incorporating predictive models of the system's future states. This inclusion of MPC in the conversation underscores the evolving landscape of control systems and their pivotal role in advancing automation and system optimization across various fields.

II. PID CONTROLLER

Control systems are ubiquitously integrated into various aspects of our daily lives, orchestrating a multitude of operations seamlessly that their pervasive influence often goes unnoticed. These systems are instrumental in automating tasks, enhancing efficiency, and alleviating the need for human intervention in routine processes. Within residential settings, control systems are employed in devices such as thermostats to maintain optimal ambient

temperatures, refrigeration units to preserve food at specific temperatures, and computing devices to regulate power distribution among electrical components. Similarly, automotive technologies leverage control systems for engine management, climate regulation, and implementing advanced features like cruise control, which transfers the task of maintaining vehicle speed from the driver to an automated system [1][2].

This narrative extends into the professional domain, where control systems play a critical role in facilitating operations. Examples include elevators, which exemplify the application of control systems in vertical transportation, automatic doors enhancing accessibility and security, and complex scenarios to control over the storage. Additionally, control systems find extensive application in automotive research and development [3], notably in the Advanced Driver Assistance Systems (ADAS) and Autonomous Driving (AD).

Municipal and infrastructural applications further demonstrate the versatility of control systems, ranging from traffic management to the regulation of utilities, exemplifying their contribution to urban planning and public welfare. The inception of control systems can be traced back to the invention of the flyball governor by James Watt [4]. This mechanism, designed to regulate the power output of steam engines, marks a pivotal advancement in automation technology. By linking the movement of flyballs, propelled by centrifugal force, to a throttle valve, the flyball governor autonomously adjusted steam flow, thereby stabilizing engine output. This innovation laid the groundwork for subsequent developments in control systems, illustrating the principle of feedback control [5].

Advancements in control technology have since facilitated the creation of supermaneuverable aircraft, which rely on intricate control systems to execute maneuvers beyond the capabilities of human pilots alone. Such developments underscore the role of control systems in augmenting human capabilities, a trend that is progressively shifting towards the development of fully autonomous systems [6]. The emergence of autonomous vehicles and robotic manufacturing underscores a paradigm shift, where control systems no longer supplement human efforts but are poised to assume complete operational control [7][8].

Reflecting on the evolution of control systems from mechanical governors to sophisticated autonomous technologies, it is evident that these systems have transcended their initial applications, becoming integral to the advancement of modern technology. As we continue to explore and innovate within this domain, the potential for control systems to redefine our interaction with technology and the environment remains boundless, heralding a future where automation and efficiency are paramount.

A. Foundations of Control Systems: From Open-Loop to Closed-Loop Control

To grasp the nuances of closed-loop control systems, it is imperative to first comprehend the foundational principles of open-loop control [9]. Utilizing the act of adjusting a shower's temperature as an illustrative example, this discussion delineates the interaction between a controller and

the system it aims to regulate, referred to in this context as the "plant." The controller, in this scenario embodied by the individual adjusting the shower, manipulates the plant (the shower mechanism from knob to nozzle) to achieve a desired outcome, which is the water temperature. In control system design, the terminology of input and output is crucial, with the understanding that one entity's output becomes another's input. This dynamic is pivotal in differentiating between open-loop and closed-loop control systems. In open-loop control, such as the initial adjustment of a shower's temperature, the controller acts based on predetermined knowledge or a reference point without considering the system's output. This lack of feedback from the plant's output characterizes open-loop control's inherent limitation [10].

Conversely, closed-loop control incorporates feedback from the system's output into the decision-making process of the controller. This is exemplified in the shower temperature adjustment scenario, where after the initial setting, the individual uses their hand to gauge the water temperature. This feedback, or observed output, is then used to make further adjustments to the plant via the controller, minimizing the error between the desired temperature (reference) and the actual temperature (output). This iterative process of adjustment based on feedback exemplifies the essence of closed-loop control, aiming for a state where the error is nullified, and the system's output aligns with the desired reference. The concept of feedback loops is further explored through positive and negative feedback mechanisms. Positive feedback loops, such as Google's search algorithm improvement cycle or the exacerbating cycle of global warming, illustrate scenarios where the system's output reinforces the input in a self-amplifying loop [11]. While beneficial in certain contexts, positive feedback can lead to instability within control systems. Thus, for the purpose of achieving stable system behavior, negative feedback loops are preferred. In these loops, the system's output is subtracted from the reference to continually adjust and refine the control action, thereby stabilizing the system's performance. Closed-loop control's application extends beyond simple mechanical adjustments to complex systems like elevators, where the control code (controller) interacts with the elevator mechanism (plant) to adjust its position based on feedback from height sensors. This feedback informs the controller of the discrepancy between the elevator's current position and the desired floor (reference), enabling precise adjustments to be made through proportional control and other control strategies [12]. This foundational understanding sets the stage for a deeper exploration of proportional control and its role in enhancing the precision and stability of control systems.

B. Proportional Control

In the realm of control systems, understanding the distinction between closed-loop and open-loop control is fundamental. The initial action of adjusting a shower's temperature serves as an illustrative example of both control paradigms. Initially, the act of setting the shower temperature represents open-loop control, where the adjustment is made based on pre-existing knowledge without feedback from the system's output. This scenario transitions into a closed-loop control when the individual samples the water's temperature with their hand and makes subsequent adjustments based on this feedback.

The design of a controller revolves around formulating its response to a defined error, essentially determining the appropriate output for given error conditions. In the domain of proportional control, the controller's output is devised to be directly proportional to the error magnitude. This relationship is characterized by the equation $Output = Error \times K_p$, where K_p represents the proportional gain—a predefined constant that influences the controller's sensitivity to error.

Applying this concept to the shower example, the magnitude of adjustment to the shower's temperature control knob is proportional to the deviation of the current temperature from the desired temperature. If the temperature is significantly higher or lower than desired, a substantial adjustment is necessary. Conversely, if the deviation is minor, only a slight adjustment is warranted. This intuitively aligns with the principle that the effort to correct an error [13] should correspond to the error's size.

The proportional gain, K_p , plays a pivotal role in determining the controller's behavior, with its value affecting the system's response to errors. This gain can be positive or negative, contingent upon the relationship between the controller's output and the system's error. An incorrect sign of K_p can result in the system acting in opposition to the intended direction of correction. Determining the optimal value for K_p is a nuanced process, requiring experimentation and adjustment to ascertain how the system responds to different gain settings [14].

For instance, in an elevator control scenario, if the goal is to achieve a maximum acceleration of 2 meters per second squared when the elevator is 20 meters away from the desired floor, a proportional gain of 0.1 might be appropriate. This setting implies that an error of 20 meters, when multiplied by a K_p of 0.1, yields a controller output that results in the desired acceleration.

The creation of a proportional controller is a step towards developing a comprehensive control system capable of addressing more complex control tasks. Such a controller calculates the error as the difference between the current state and the desired state, and then applies the proportional gain to determine the control action [15]. This straightforward mechanism underscores the effectiveness of proportional control in regulating system behaviors, making it a fundamental strategy in control system design.

C. Derivative Control

In this discussion, we delve into the concept of derivative control within control systems, elucidating its definition, benefits, and implementation methodology. The discussion begins by addressing a common issue encountered in control systems, particularly when a controller continuously accelerates a system towards a setpoint without considering the velocity at which the setpoint is approached. This is a fundamental characteristic of systems governed solely by proportional control, where the controller's output is directly proportional to the error, formulated as

$Output = Error \times K_p$, and ceases acceleration only when the error reaches zero [16][17].

The consequence of such unmodulated acceleration is the system attaining excessive velocity as it nears the setpoint, leading to overshoots. To mitigate this, the introduction of deceleration, or negative acceleration, is necessary well before the setpoint is reached, aiming to reduce the system's velocity to near-zero as it approaches the target. This is where derivative control plays a pivotal role.

Derivative control focuses on the rate of change of the error, providing a mechanism to anticipate system behavior and adjust the control action accordingly. Mathematically, if a function $y = f(x)$ represents the system, its derivative $f'(x)$ at any point signifies the slope or the rate of change at that point. Applied to control systems, if $x = f(t)$ represents the distance from a goal over time t , the derivative $x' = \frac{df(t)}{dt}$ corresponds to the velocity or the rate at which the distance changes [18].

In practical terms, considering discrete measurements as opposed to continuous functions, the derivative or rate of change between two points is calculated using the rise-over-run method, where the rise represents the change in distance (Δx) and the run represents the change in time (Δt). This yields the derivative $x' = \frac{\Delta x}{\Delta t}$, which in the context of a control system, translates to the velocity.

Utilizing derivative control involves manipulating the system's control output based on the derivative of the error, allowing for adjustments before the error itself becomes zero. This is crucial for smoothly reaching and maintaining a setpoint without overshooting [19]. For example, in a vehicle lane-changing scenario, derivative control would modulate the control output not just based on the lateral distance to the desired lane (error) but also based on the rate at which this distance is decreasing (derivative of the error). This ensures that as the vehicle approaches the desired lane, adjustments become finer and the transition smoother.

The implementation of derivative control in a control system is achieved by adding a term proportional to the derivative of the error to the controller's output. This can be represented as $Output = K_p \cdot Error + K_d \cdot \frac{d(Error)}{dt}$, where

K_d is the derivative gain, a constant that determines the system's sensitivity to the rate of change of the error [20]. The inclusion of this term allows the controller to counteract rapid changes in error, effectively slowing down the system as it approaches the setpoint and reducing overshoot.

Overall, derivative control enhances the capability of control systems by providing a means to anticipate future system states and adjust control actions preemptively. This is particularly valuable in applications requiring precise control and stabilization as systems approach their target states, exemplifying the importance of considering both the

magnitude and the rate of change of the error in control system design.

D. Integral Control

Integral control represents a critical component in the development of comprehensive control strategies, particularly within the context of Proportional-Derivative-Integral (PDI) controllers. This discussion elucidates the fundamental principles underlying integral control, its necessity, and its application in resolving specific control challenges not addressable through proportional or derivative control alone.

One common scenario illustrating the limitations of Proportional-Derivative (PD) control involves the addition of variable loads to a system, such as people entering an elevator. Observations may reveal that upon integrating such dynamic variables, the PD controller's efficacy diminishes, notably failing to maintain the elevator at the designated setpoint due to the unbalanced forces introduced by the additional weight. This deviation cannot be rectified through adjustments in output gain or by the introduction of frictional components; it stems from an intrinsic inadequacy in the PD control model to account for continuous, accumulative errors over time [21].

The essence of the problem lies in the static nature of PD control when facing persistent, unidirectional biases in the system—such as the added weight in an elevator scenario. With a purely PD-controlled system, once equilibrium is achieved (where the forces exerted by the motor and counterweight balance the combined force of the elevator and its additional load), the elevator may stabilize at a position below the intended setpoint due to the controller's inability to compensate for the continuous bias [22].

Integral control addresses this challenge by integrating the error over time, thereby accumulating the historical errors. Mathematically, if the error function over time is denoted as $e(t)$, the integral control action $I(t)$ can be expressed as:

$$I(t) = K_i \int_0^t e(\tau) d\tau \quad (1)$$

where K_i is the integral gain, and $e(\tau)$ represents the error at time τ . This approach ensures that even minimal, persistent errors contribute to the control action [23], effectively correcting for biases that a PD controller overlooks.

The integration of errors over time ensures that the control action is not solely reactive but also compensatory, accounting for accumulated discrepancies between the system's actual state and the setpoint. For instance, in the elevator example, as people add weight, the integral component of the controller accumulates this error, gradually adjusting the control output to return the elevator to its setpoint despite the additional load.

However, care must be taken to manage the integral component effectively. Unchecked integration can lead to "integral windup," a condition where the integral term becomes excessively large during extended periods of error,

potentially leading to unstable control actions. To mitigate this, implementing a saturation limit on the integral term is advisable, ensuring that it remains within a manageable range that reflects the physical constraints and capabilities of the system [24][25].

To sum up, integral control provides a mechanism to address biases and persistent errors within a system, complementing the instantaneous error correction offered by proportional control and the anticipatory adjustments afforded by derivative control. By summing the errors over time, integral control ensures that even small, continuous discrepancies are accounted for, enabling the system to achieve and maintain its setpoint under a broader range of conditions. This comprehensive approach to control, encapsulating proportional, integral, and derivative actions, equips engineers with a versatile toolset for designing robust, responsive control systems capable of handling complex, dynamic challenges.

III. MODEL PREDICTIVE CONTROL

Model Predictive Control (MPC) represents a paradigm shift in the approach to control system design, offering a framework that blends predictive modeling with optimization-based control strategies. Through its application across various domains, MPC has proven to be a versatile and effective tool for addressing complex engineering challenges [26]. This document has endeavored to illuminate the foundational principles of MPC, bridging the gap between abstract theoretical concepts and their practical applications in real-world scenarios.

A. The Emergence of Model Predictive Control

In recent decades, the engineering discipline has witnessed the advent of a transformative computational methodology known as Model Predictive Control (MPC), which has garnered widespread acclaim for its versatility and applicability across diverse systems [27]. This technique stands out for its unique approach to anticipating future system behaviors to make informed current decisions. Despite its ubiquity and potential, the underlying principles and operational intricacies of MPC remain elusive to a considerable segment of the engineering community. This complexity is often attributed to its abstract nature and the initial perception of non-intuitiveness. However, these narrative endeavors to demystify MPC, presenting it as an accessible, intuitively aligned tool that integrates seamlessly with real-world applications [28]. Mastery in engineering problem-solving mandates proficiency in three fundamental competencies: intuitive problem understanding, mathematical quantification, and the translation of these elements into computational algorithms. This document aims to cultivate these skills, offering a comprehensive exploration of MPC within the context of engineering problem-solving.

B. Fundamentals of Model Predictive Control

Before delving into the mathematical formalism of MPC, it is imperative to establish an intuitive grasp of its functionality. Consider the example of a spacecraft lander tasked with achieving a smooth descent onto a planetary surface. This scenario encapsulates the essence of MPC, where the primary objective is to modulate the spacecraft's thrust to counteract gravitational forces, thereby ensuring a

controlled landing. The distinction between a successful soft landing and a catastrophic impact can be illustrated through the trajectory of the lander's altitude over time, highlighting the critical role of MPC in navigating such dynamic challenges.

MPC operates on the principle of optimizing control actions over a finite prediction horizon based on a predictive model of the system. The core objective is to minimize a cost function that typically includes terms for tracking errors and control efforts [29]. Formally, the predictive model can be represented as:

$$x(t+1) = f(x(t), u(t)) \quad (2)$$

where $x(t)$ denotes the system state at time t , $u(t)$ represents the control input, and f is the function describing the system dynamics. The cost function J to be minimized over the prediction horizon N is defined as:

$$J = \sum_{k=0}^{N-1} \left(\|y(k|t) - y_{ref}(k|t)\|_Q^2 + \|u(k|t)\|_R^2 \right) \quad (3)$$

where $y(k|t)$ is the predicted output at time k based on information available at time t , y_{ref} is the reference trajectory, and Q and R are weighting matrices that prioritize tracking performance and control effort, respectively [30].

To elucidate the principles of MPC, consider the spacecraft lander scenario where the goal is to minimize the vertical descent velocity to achieve a soft landing. The system model includes forces such as gravity and thrust from the spacecraft's engines. By applying MPC, the control algorithm computes the optimal thrust adjustments at discrete time intervals, considering both the current state and predicted future states of the lander. The prediction horizon in MPC facilitates the anticipation of future trajectory adjustments, allowing for the calculation of control inputs that navigate the lander towards a target state, ideally with zero vertical velocity upon touchdown.

C. Implementing MPC in Engineering Applications

Translating the theoretical constructs of MPC into practical applications necessitates a comprehensive understanding of both the system under consideration and the computational tools at one's disposal. Implementing MPC involves developing a detailed mathematical model of the system, selecting appropriate parameters for the prediction horizon and cost function, and employing numerical optimization techniques to solve the control problem.

Effective implementation of MPC requires addressing computational challenges, particularly regarding the real-time execution of optimization algorithms. Advances in computational hardware and software have significantly mitigated these challenges, enabling the application of MPC in real-time scenarios such as autonomous vehicles, industrial process control, and aerospace navigation [31][32].

IV. COMPARATIVE ANALYSIS OF PID CONTROLLERS AND MODEL PREDICTIVE CONTROL THROUGH SIMULATION

This chapter employs a structured approach to compare PID controllers and MPC, utilizing simulation as a tool to replicate realistic control scenarios in a controlled, repeatable manner. The simulations are designed to test the controllers'

responses to a range of challenges, including setpoint changes, load disturbances, and parameter variations. By meticulously analyzing the controllers' performance in these scenarios, this chapter aims to uncover the relative advantages and disadvantages of each control strategy, providing a comprehensive resource for control system designers and researchers.

In the following sections, we will delve into the systematic exploration. The chapter aspires to contribute to the body of knowledge in control engineering, facilitating informed decision-making in the design and implementation of control systems.

A. Mitigating Integrator Wind-Up in PID Control Systems: Strategies and Implementation Insights

In the current discourse, we delve into the intricacies of control system design, with a particular emphasis on the mitigation of integrator wind-up phenomena within Proportional-Integral-Derivative (PID) controllers. This exploration is instantiated through a detailed analysis of a specific implementation scenario, whereby the control system is tasked with managing the stability of an elevator under varying load conditions.

The foundational concept underpinning this analysis is the PID control algorithm, a cornerstone in the domain of automatic control systems, described by the equation:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt} \quad (4)$$

where $u(t)$ represents the control signal output at time t , $e(t)$ denotes the error between the desired setpoint and the measured process variable, and K_p , K_i , and K_d are the proportional, integral, and derivative gains, respectively [33].

In this analysis, the focal point is the integral component of the PID controller, particularly its susceptibility to an anomaly known as integrator wind-up. This phenomenon occurs when the integrator's output exceeds the bounds of the system's actuator capabilities, resulting in a delay in system response and potential stability issues [34]. To counteract this, a technique known as integrator clamping is employed, setting upper and lower limits on the integrator's output to ensure it remains within a feasible operational range. The implementation of this safeguard is crucial for the stability and responsiveness of the control system, especially in scenarios involving significant load variations, such as the elevator model discussed herein.

The specific parameters of the implemented control system include a maximum integrator wind-up limit of 5 and an output ceiling of 2.5. These parameters were meticulously chosen to balance the need for preventing integrator wind-up and maintaining control system performance. Experimental observations of the system, under a load mass of 200 units, revealed a generally smooth response. However, a slight delay was noted in achieving the final setpoint adjustment, attributed to the limitations imposed by the integrator wind-up prevention mechanism.

Further insights were gleaned through the activation of a debug mode, revealing that the integral component reached

its imposed maximum value, yet requiring further adjustment to adequately counterbalance the applied load. This scenario underscores the inherent trade-off between integrator wind-up prevention and the dynamic responsiveness of the control system. The integral component's inability to decrease in the absence of a negative error (i.e., when the system's output exceeds the setpoint) necessitates a transient overshoot for its reduction, illustrating a critical aspect of control system design—managing the interplay between accuracy and stability [35].

To extend the analysis, the control system's performance was evaluated under reverse loading conditions, simulating a negative mass scenario. This test further illuminated the challenges posed by integrator wind-up, as the control system exhibited greater difficulty in adjusting the integral component from positive to negative values, compared to the initial scenario. This observation suggests that prior knowledge of the system's operational parameters, such as consistent counterweight and load mass relationships, could inform more nuanced integrator clamping strategies, potentially enhancing system performance.

The discourse concludes with a reflection on the control system's implementation from a software engineering perspective. Despite the apparent simplicity of the control algorithm in code form, the underlying logic and theoretical principles are complex and crucial for the effective application of PID controllers across diverse scenarios. This analysis not only elucidates specific strategies for mitigating integrator wind-up but also highlights the broader importance of a deep understanding of control theory principles in designing robust and responsive automatic control systems.

B. Bridging Mathematical Concepts with Real-World Applications in Vehicle Dynamics

In this comprehensive discourse, the objective is to elucidate the transition from theoretical mathematical foundations to their practical embodiment within computer code, focusing on the simulation of vehicular dynamics. This narrative is intended to serve as an exposition of the pivotal linkage between abstract theoretical concepts and their tangible applications in real-world scenarios, as demonstrated through a simulation environment.

The subject of this analysis is the bicycle model simulation, a fundamental representation in vehicle dynamics, illustrating the behavior of a vehicle (in this instance, modeled as a bicycle for simplification) as it navigates through a predefined path. This simulation is delineated over a duration of ten seconds and is visualized in a dual subplot format wherein the first subplot presents a straight trajectory segmented into five distinct lanes. Within this visual representation, the bicycle's trajectory is demarcated by a red line, indicating its actual path, whereas a blue line signifies the reference trajectory that the vehicle is intended to follow. This reference trajectory is established independently of the control system, implying that the vehicle is not inherently programmed to adhere to this path but rather relies on the control system to compute the necessary adjustments in steering to conform to the trajectory.

A notable feature within this simulation is represented by a magenta line, positioned anterior to the bicycle model, which depicts the projected trajectory based on the vehicle's current state and anticipated system inputs. This projection is rooted in the optimization of a predefined cost function, facilitated by the application of augmented A and B matrices within the control algorithm, thereby allowing for the foresight of vehicle positioning over a future horizon. Initially, this horizon spanned five samples; however, for the purposes of this simulation, it has been extended to encompass twenty samples, thereby providing a more extensive preview of the vehicle's trajectory and enabling more precise control interventions.

The augmentation from a five-sample horizon to a twenty-sample horizon necessitates a corresponding expansion in the dimensionality of the matrices involved, transforming a 5x5 matrix to a 20x20 matrix to accommodate the extended predictive scope. Concurrently, the simulation incorporates a finer temporal resolution, with the sampling interval reduced from 0.1 seconds to 0.02 seconds. This adjustment results in a smoother simulation experience by increasing the number of samples processed within the ten-second timeframe, thus offering a more granular view of the vehicle's dynamics.

Further examination of the bicycle model reveals the intricate relationship between the vehicle's body orientation, the steering angle (Δ), and the yaw angle (ψ), which collectively govern the vehicle's navigational adjustments. The steering angle (Δ) is measured relative to the vehicle's body, while the yaw angle (ψ) is defined with respect to an inertial reference frame, illustrating the vehicle's angular orientation as it executes a lane-change maneuver [36].

The empirical observation of this simulation yields three distinct plots as functions of time, showcasing the evolution of the steering wheel angle, the yaw angle, and the lateral displacement (Y values) of the bicycle. These plots reveal the controller's efficacy in tracking the predefined reference values for each parameter, highlighting the dynamic adaptability of the control system in real-time.

This exploration culminates in a preparatory segue into a detailed code walkthrough, intended to bridge the gap between the theoretical underpinnings of control systems and their practical implementation in simulating complex dynamic systems. Through this analysis, the intricate ballet of mathematical theory, control system design, and computational modeling is unraveled, shedding light on the sophisticated interplay between theoretical constructs and their practical applications in engineering and computational simulation.

V. SUMMARY

In this comprehensive discussion, the focus was placed on the evolution of control system design, moving from conventional Proportional-Integral-Derivative (PID) frameworks to the incorporation of advanced control strategies capable of addressing the limitations inherent in PID controllers. Central to this discussion was the delineation of the reactive nature of PID controllers, which, by design,

respond based on historical data without foresight into future system states or the ability to integrate constraints smoothly. This limitation becomes particularly evident in scenarios requiring the control system to avoid overshooting a target or to maintain output within specified bounds, wherein PID controllers might falter due to their inability to plan or utilize advanced process knowledge.

The exploration extended into the realm of advanced control strategies, notably Model Predictive Control (MPC) and Nonlinear Control, which offer predictive capabilities and adaptability to non-linear dynamics, respectively. MPC is highlighted for its utilization of a physics-based model to anticipate the future states of a system, thereby facilitating informed decision-making that optimizes current and future system performance while adeptly managing constraints. This predictive prowess necessitates a more sophisticated reference model, employing a cost function that balances multiple objectives, such as minimizing distance from a target while considering factors like control smoothness and acceleration, to ensure an optimized and smooth response. Conversely, Nonlinear Control Systems are underscored for their capacity to handle systems where the relationship between controller output and system response is not linear, showcasing their utility in a broader spectrum of applications beyond the linear approximations typically associated with PID control.

Additionally, the discourse provided practical implementation advice for the individual components of PID controllers, underscoring the selective application of P, I, and D components based on the specific requirements and dynamics of the system under control. This included insights into the necessity of the integral component for systems requiring constant input to maintain stability, the use of the derivative component to preemptively dampen system response, and strategies to mitigate issues such as integrator windup and noise susceptibility inherent in PID components.

VI. CONCLUSION

The exploration into advanced control strategies illuminates the path forward in control system design, advocating for a shift towards methodologies that embrace predictive capabilities and can adeptly navigate the intricacies of nonlinear system dynamics. Through the lens of Model Predictive Control and Nonlinear Control Systems, this discussion elucidates the limitations of traditional PID controllers and posits these advanced strategies as essential for tackling the dynamic challenges faced in contemporary engineering applications. This shift not only augments the ability to manage complex systems with greater efficacy but also broadens the applicability of control systems across a diverse array of scenarios, marking a significant evolution in the field of automatic control.

Furthermore, the practical insights offered into the nuanced application of PID controller components underscore the importance of a tailored approach in control system design. By discerning the specific needs of a system and the environmental factors at play, these insights equip practitioners with the knowledge to optimize control system performance effectively.

In summary, the transition from PID to advanced control strategies represents a critical advancement in the field of control systems, promising enhanced adaptability, efficiency, and precision in tackling the multifaceted challenges of modern systems. This progression not only signifies a notable shift towards more intelligent control mechanisms but also highlights the pivotal role of advanced control strategies in shaping the future of automation and system dynamics.

REFERENCES

- [1] GL Zhang, "Research on PID parameter tuning based on intelligent fusion optimization algorithm in control system", *Int J u-ande-ServSci Technol*, 2016.
- [2] Y. Wang and S. Boyd, "Fast Model Predictive Control Using Online Optimization", *IEEE Transactions on Control Systems Technology*, vol. 18, no. 2, pp. 267-278, 2010.
- [3] S. Trimboli, S. Di Cairano, A. Bemporad, and I. Kolmanovsky, "Model predictive control for automotive time-delay processes: an application to air-to-fuel ratio," in *Proc. 8th IFAC Workshop on Time-delay Systems*, Sinaia, Romania, 2009, pp. 1-6.
- [4] R. Mehra, J. Amin, J. Hedrick, C. Osorio, and S. Gopalasamy, "Active suspension using preview information and model predictive control," in *Proc. IEEE Int. Conf. Control Applications*, Hartford, CT, 1997, pp. 860-865.
- [5] S. Di Cairano, D. Yanakiev, A. Bemporad, I. Kolmanovsky, and D. Hrovat, "An MPC design flow for automotive control and applications to idle speed regulation," in *Proc. 47th IEEE Conf. on Dec. and Control*, 2008, pp. 5686-5691.
- [6] D. Hrovat, "MPC-based idle speed control for IC engine," in *Proc. of FISITA conference*, Prague, Czech Rep., 1996.
- [7] Z. Šehić and Z. Šehić-Galić, "Comparison Study for PID and MPC Controller based on Orthonormal Functions", *Advanced Research in Scientific Areas*, pp. 1700-1704, 2012.
- [8] Veysi, P.; Adeli, M.; Peirov Naziri, N. Development and Validation of an Artificial Neural Network for the Recognition of Custom Dataset with YOLOv4. https://www.researchgate.net/publication/379366046_Development_and_Validation_of_an_Artificial_Neural_Network_for_the_Recognition_of_Custom_Dataset_with_YOLOv4
- [9] N. Giorgetti, G. Ripaccioli, A. Bemporad, I. Kolmanovsky, and D. Hrovat, "Hybrid Model Predictive Control of Direct Injection Stratified Charge Engines," *IEEE/ASME Transactions on Mechatronics*, vol. 11, no. 5, pp. 499-506, 2006.
- [10] Bennett Stuart, "A history of control engineering 1930-1955", *IET*, pp. 48, 1993.
- [11] Ahmed A. Abdelrauf, M. Abdel-Gelil and E. Zakzouk, "Adaptive PID controller based on model predictive control", *Control Conference (ECC) 2016 European*, pp. 746-751, 2016.
- [12] Veysi, P.; Adeli, M.; Peirov Naziri, N. A Gentle Approach to Multi-Sensor Fusion Data Using Linear Kalman Filter. https://www.researchgate.net/publication/379836342_A_Gentle_Approach_to_Multi-Sensor_Fusion_Data_Using_Linear_Kalman_Filter
- [13] G. Stewart and F. Borrelli, "A Model Predictive Control Framework for Industrial Turbodiesel Engine Control," in *Proc. 47th IEEE Conf. on Dec. and Control*, Cancun, Mexico, Dec 2008, pp. 5704-5711.
- [14] Veysi, P.; Adeli, M.; Peirov Naziri, N. Implementation of Kalman Filtering and Multi-Sensor Fusion Data for Autonomous Driving. https://www.researchgate.net/publication/379908832_Implementation_of_Kalman_Filtering_and_Multi-Sensor_Fusion_Data_for_Autonomous_Driving
- [15] R. Schalllock, K. Muske, and J. Peyton Jones, "Model predictive functional control for an automotive three-way catalyst," *SAE Int. J. Fuels and Lubricants*, vol. 2, no. 1, pp. 242-249, 2009.
- [16] G. W. Juette and L. E. Zeffanella, "Radio noise currents in short sections on bundle conductors (Presented Conference Paper style)," presented at the *IEEE Summer power Meeting*, Dallas, TX, June 22-27, 1990, Paper 90 SM 690-0 PWRs.
- [17] G. Ripaccioli, A. Bemporad, F. Assadian, C. Dextreit, S. Di Cairano, and I. Kolmanovsky, "Hybrid Modeling, Identification, and Predictive Control: An Application to Hybrid Electric Vehicle Energy

- Management," in *Hybrid Systems: Computation and Control*, ser. Lec. Not. in Computer Science. Springer, 2009, vol. 5469, pp. 321-335.
- [18] S. Di Cairano, H. Tseng, D. Bernardini, and A. Bemporad, "Steering vehicle control by switched model predictive control," in *6th IFAC Symp. on Advances in Automotive Control*, Munich, Germany, 2010.
 - [19] N. Giorgetti, A. Bemporad, E. H. Tseng, and D. Hrovat, "Hybrid model predictive control application towards optimal semi-active suspension," *Int. J. Control*, vol. 79, no. 5, pp. 521-533, 2006.
 - [20] C. Vermillion, J. Sun, and K. Butts, "Predictive control allocation for a thermal management system based on an inner loop reference model: Design, analysis, and experimental results," *IEEE Trans. Contr. Systems Technology*, vol. 19, no. 4, pp. 772-781, 2011.
 - [21] S. Di Cairano, D. Yanakiev, A. Bemporad, I. V. Kolmanovsky, and D. Hrovat, "Model predictive idle speed control: Design, analysis, and experimental evaluation," *IEEE Trans. Contr. Systems Technology*, vol. 20, no. 1, pp. 84-97, 2012.
 - [22] S. Di Cairano and A. Bemporad, "Model predictive control tuning by controller matching," *IEEE Trans. Automatic Control*, vol. 55, no. 1, pp. 185-190, jan. 2010.
 - [23] C. Garcia, D. Prett, and M. Morari, "Model predictive control: theory and practice-a survey," *Automatica(Oxford)*, vol. 25, no. 3, pp. 335-348, 1989.
 - [24] S. Di Cairano and H. Tseng, "Driver-assist steering by active front steering and differential braking: Design, implementation and experimental evaluation of a switched model predictive control approach," in *Proc. 49th IEEE Conf. on Dec. and Control*, dec. 2010, pp. 2886-2891.
 - [25] N. Giorgetti, G. Ripaccioli, A. Bemporad, I. Kolmanovsky, and D. Hrovat, "Hybrid Model Predictive Control of Direct Injection Stratified Charge Engines," *IEEE/ASME Transactions on Mechatronics*, vol. 11, no. 5, pp. 499-506, 2006.
 - [26] R. Amari, M. Alamir, and P. Tona, "Unified MPC strategy for idlespeed control, vehicle start-up and gearing applied to an Automated Manual Transmission," in *Proc. 17th IFAC World Congress*, Seoul, South Korea, 2008, pp. 7079-7085.
 - [27] S. Qin and T. Badgwell, "A survey of industrial model predictive control technology," *Control Engineering Practice*, vol. 93, no. 316, pp. 733-764, 2003.
 - [28] P. Falcone, F. Borrelli, J. Asgari, H. Tseng, and D. Hrovat, "Predictive active steering control for autonomous vehicle systems," *IEEE Trans. Contr. Systems Technology*, vol. 15, no. 3, pp. 566-580, 2007.
 - [29] P. Ortner and L. del Re, "Predictive Control of a Diesel Engine Air Path," *IEEE Trans. Contr. Systems Technology*, vol. 15, no. 3, pp. 449-456, 2007.
 - [30] T. Hatanaka, T. Yamada, M. Fujita, S. Morimoto, and M. Okamoto, "Explicit receding horizon control of automobiles with continuously variable transmissions," *Nonlinear Model Predictive Control*, vol. 384, pp. 561-569, 2009.
 - [31] S. Di Cairano, A. Bemporad, I. Kolmanovsky, and D. Hrovat, "Model predictive control of magnetically actuated mass spring dampers for automotive applications," *Int. J. Control*, vol. 80, no. 11, pp. 1701-1716, 2007.
 - [32] M. Canale, M. Milanese, and C. Novara, "Semi-active suspension control using fast model-predictive techniques," *IEEE Trans. Contr. Systems Technology*, vol. 14, no. 6, pp. 1034-1046, 2006.
 - [33] H. Borhan, C. Zhang, A. Vahidi, A. Phillips, M. Kuang, and S. Di Cairano, "Predictive energy management of a power-split hybrid electric vehicle," in *Proc. 49th IEEE Conf. on Dec. and Control*, Atlanta, GA, 2010, pp. 4890-4895.
 - [34] P. Falcone, H. Tseng, F. Borrelli, J. Asgari, and D. Hrovat, "MPCbased yaw and lateral stabilisation via active front steering and braking," *Vehicle System Dynamics*, vol. 46, no. Supplement, pp. 611-628, 2008.
 - [35] A. Bemporad, M. Morari, V. Dua, and E. Pistikopoulos, "The Explicit Linear Quadratic Regulator for Constrained Systems," *Automatica*, vol. 38, no. 1, pp. 3-20, 2002.
 - [36] M. Johansson and A. Rantzer, "Computation of piece-wise quadratic Lyapunov functions for hybrid systems," *IEEE Trans. Automatic Control*, vol. 43, no. 4, pp. 555-559, 1998.